# PP-DocLayout: A Unified Document Layout Detection Model to Accelerate Large-Scale Data Construction

Ting Sun, Cheng Cui, Yuning Du, Yi Liu
PaddlePaddle Team, Baidu Inc.
{sunting13, cuicheng01} @baidu.com



## Abstract

*Document layout analysis is a critical preprocessing step in document intelligence, enabling the detection and localization of structural elements such as titles, text blocks, tables, and formulas. Despite its importance, existing layout detection models face significant challenges in generalizing across diverse document types, handling complex layouts, and achieving real-time performance for large-scale data processing. To address these limitations, we present PP-DocLayout, which achieves high precision and efficiency in recognizing 23 types of layout regions across diverse document formats. To meet different needs, we offer three models of varying scales.* ***PP-DocLayout-L*** *is a high-precision model based on the RT-DETR-L detector, achieving 90.4% mAP@0.5 and an end-to-end inference time of 13.4 ms per page on a T4 GPU.* ***PP-DocLayout-M*** *is a balanced model, offering 75.2% mAP@0.5 with an inference time of 12.7 ms per page on a T4 GPU.* ***PP-DocLayout-S*** *is a high-efficiency model designed for resource-constrained environments and real-time applications, with an inference time of 8.1 ms per page on a T4 GPU and 14.5 ms on a CPU. This work not only advances the state of the art in document layout analysis but also provides a robust solution for constructing high-quality training data, enabling advancements in document intelligence and multimodal AI systems. Code and models are available at* `https://github.com/PaddlePaddle/PaddleX`.

## 1. Introduction

The rapid development of large language models (LLMs) and multi-modal document understanding systems [10, 11] has led to a significant increase in the demand for high-quality structured training data. Document layout detection, which identifies and localizes structural elements (e.g., text blocks, tables, and figures), plays a pivotal role in transforming raw document images into machine-readable formats. As illustrated in **Figure 1**, layout detection is the foundational step for a variety of downstream tasks, including table recognition, formula recognition, OCR and information extract. For instance, in the case of table recognition, layout detection models accurately locate and define the boundaries of tables within document images, enabling the extraction of table regions for further processing, such as parsing the tabular structure and extracting the underlying data. This structured table data is extremely valuable for applications ranging from data analysis to information retrieval. Similarly, for formula recognition, layout detection models detect and localize formula regions within documents. This allows for the extraction of these regions, which can then be fed into specialized formula recognition systems. The resulting structured formula data not only enhances the machine's understanding of mathematical content but also enriches training datasets, improving models' ability to recognize and interpret formulas in various contexts.

However, despite its potential, existing layout detection models face three critical limitations. (1) Poor generalization across document types. Current approaches predominantly focus on academic papers, resulting in poor performance on other document types such as magazines, newspapers, and financial reports. (2) Inadequate handling of complex layouts. The lack of comprehensive category definitions—such as the absence of separate labels for inline and interline mathematical formulas—necessitates auxiliary models, leading to increased complexity and reduced efficiency. (3) Insufficient processing speed for real-time applications. These challenges impede the effective use of layout detection in practical scenarios, particularly in domains where training large models requires efficient acquisition of abundant massive high-quality data.

To address these challenges, we introduce PP-DocLayout, a unified layout detection model that achieves state-of-the-art accuracy and real-time inference capabilities. PP-DocLayout achieves high-precision recognition and localization of layout regions across diverse document formats including Chinese or English academic papers, research reports, exam papers, books, newspapers, and mag-

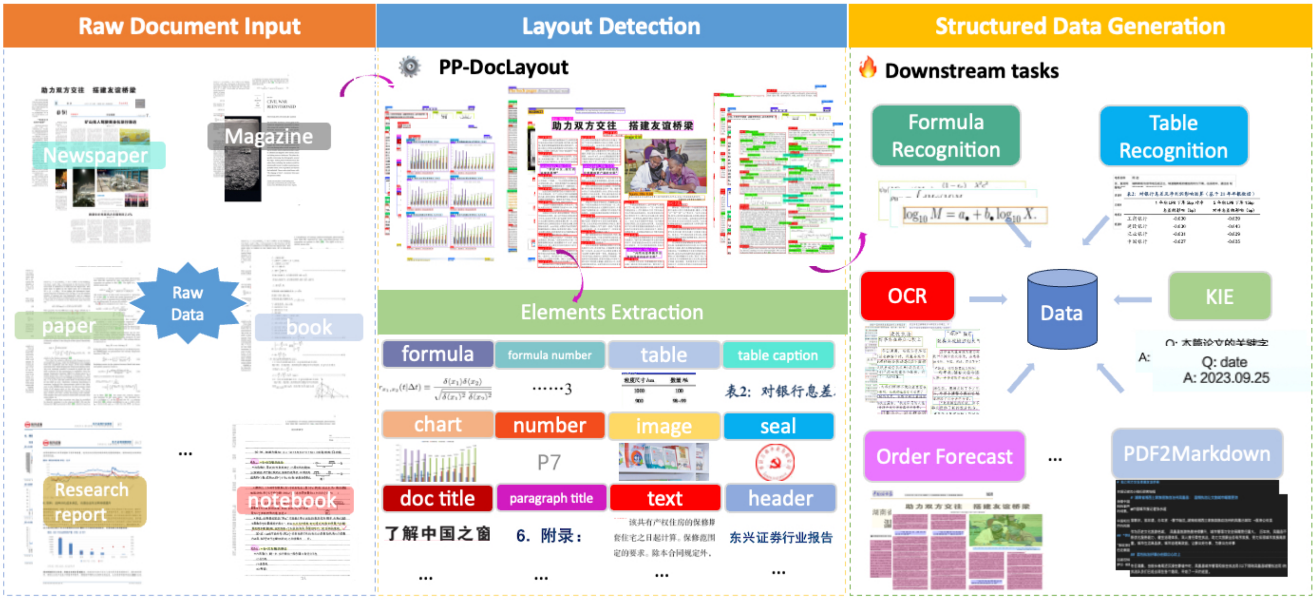


Figure 1. PP-DocLayout plays a pivotal role in layout analysis and structured data generation.

azines. This advancement significantly enhances the diversity and quality of large models training data acquisition. In addition, PP-DocLayout series supports 23 common layout categories, which cover a wide range of layout elements found in diverse documents. This clear hierarchical structure of categories facilitates improved semantic understanding and logical parsing, while the inclusion of structured data of high-value information enables more precise data processing and analysis. For addressing the critical efficiency requirements for massive-scale data construction, leveraging the high-performance PaddleX inference engine, the lightweight model demonstrates exceptional processing capabilities - handling approximately 123 pages per second on T4 GPU. These performance substantially outperforms existing open-source solutions, establishing new benchmarks for document layout analysis in terms of both accuracy and computational efficiency.

## 2. Related Work

The evolution of document layout analysis (DLA) reflects the paradigm shift from isolated component detection to holistic semantic understanding. Early unimodal methodologies framed DLA as a specialized computer vision task, adapting generic object detection frameworks (Faster R-CNN [7], YOLO [8]) with domain-specific modifications. Recently, the advanced method DocLayout-YOLO [14] ,based on YOLOv10 [8], pretrains on diverse document data and designs the GL-CRM module which obtains high-accuracy of 10 categories layout detection.

The emergence of multimodal learning has fundamentally transformed DLA methodology. The LayoutLM series [3, 10, 11] demonstrated the power of unified pre-training strategies, integrating masked visual-language modeling and spatial-aware position embeddings. Recent advancements further explore self-supervised paradigms, DiT [4] leverages massive unlabeled documents through novel pre-training objectives, and VGT [1] introduces grid-based textual encoding to preserve fine-grained typographical features. Notably, the field is witnessing convergence between DLA and document intelligence, where layout understanding serves as the foundation for higher-level semantic tasks.

Despite these advancements, several challenges remain. First, most existing methods focus on specific document types, such as academic papers, and lack generalization to diverse document categories like magazines, newspapers, and handwritten notes. Second, the detection of fine-grained elements, such as formulas, footnotes, and headers, remains under explored. Finally, the computational efficiency of layout detection methods remains a significant challenge, as many state-of-the-art models are computationally expensive and slow, limiting their applicability in real-time or large-scale document processing scenarios.

Our work addresses these limitations by proposing a unified framework for document layout detection that supports a wide range of document types and fine-grained element categories. By leveraging advanced deep learning techniques and incorporating contextual information, our method achieves robust performance across diverse layouts while maintaining computational efficiency.

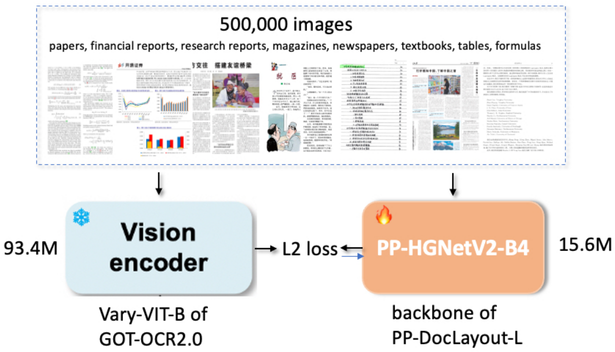


Figure 2. Knowledge Distillation Framework for the backbone of PP-DocLayout-L

## 3. Method

We introduce PP-DocLayout, a unified detection model achieving state-of-the-art performance through innovations in both data curation and algorithm design. Our approach combines three key improvement strategies.

### 3.1. Knowledge Distillation Framework

PP-DocLayout-L employs a knowledge distillation [2] paradigm to enhance the performance of document layout understanding as shown in **Figure 2**. In this framework, the visual encoder Vary-VIT-B model of GOT-OCR2.0 [9] acts as the teacher model, which is a well-trained and robust model possessing advanced document understanding capabilities. The student model, in this case, is the PP-HGNetV2-B4 backbone of PP-DocLayout-L, which is designed to learn from the teacher model.

The distillation process involves transferring knowledge from the teacher to the student by aligning their feature representations. Specifically, The teacher network parameters remain frozen while training PP-HGNetV2-B4, leveraging feature-level supervision through a fully-connected layer. Let $\mathbf{F}_{\text{Vary-VIT-B}} \in \mathbb{R}^{B\times D}$ and $\mathbf{F}_{\text{PP-HGNetV2-B4}} \in \mathbb{R}^{B\times P}$ denote the feature tensors from teacher and student networks respectively, where $D$ and $P$ represent their corresponding feature dimensions, $B$ denotes the Batch size. To bridge the dimensional discrepancy ($D \neq P$), we introduce a learnable linear projection $\varphi : \mathbb{R}^P \to \mathbb{R}^D$. The distillation loss is formulated as:

$$\mathcal{L}_{\text{Distill}} = \frac{1}{B}\sum_{i=1}^{B}\left\|\mathbf{F}^{(i)}_{\text{Vary-VIT-B}} - \varphi(\mathbf{F}^{(i)}_{\text{PP-HGNetV2-B4}})\right\|_2^2 \quad (1)$$

The distillation framework was trained on a diverse corpus containing 500000 document samples spanning five domains:

- Mathematical formulas (including equation derivations and symbolic notations)
- Financial documents (reports and balance sheets)
- Scientific literature (arxiv papers in STEM fields)
- Academic dissertations (with complex layout structures)
- Tabular data (statistical reports and spreadsheets)

Training was conducted at 768×768 resolution for 50 epochs using AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$). The distilled PP-HGNetV2-B4 achieves effective feature extraction capabilities with only 15.6 M parameters.

### 3.2. Semi-supervised Learning

In this section, we present our semi-supervised learning approach employed to enhance the performance of PP-DocLayout-M and PP-DocLayout-S models. This method leverages the high-precision capabilities of the PP-DocLayout-L model to generate pseudo-labels, which are subsequently used to augment the training data for the less complex models.

**Pseudo-Label Generation** Given an unlabeled document image $x_u$, we first generate raw predictions using the teacher model PP-DocLayout-L with trained parameters $\theta_T$ as follow:

$$P(y|x_u) = f_{\text{T}}(x_u; \theta_T) \quad (2)$$

where $P(y|x_u) \in \mathbb{R}^{N\times C}$ contains prediction scores for $N$ potential regions across $C = 23$ layout categories.

**Adaptive Threshold Selection** Traditional fixed-threshold approaches often suffer from class imbalance and difficulty levels of learning in different categories. Thus, we propose an adaptive threshold selection method to obtain high-quality pseudo-labels. Our threshold selection strategy explicitly maximizes the F-score on labeled data $x_l$ through a systematic optimization process. For each layout category $c \in \{1, ..., 23\}$, we determine the optimal threshold $\tau_c^*$ by solving:

$$\tau_c^* = \arg\max_{\tau\in[0,1]} F_c(\tau; P(y|x_l)) \quad (3)$$

where $F_c(\tau)$ denotes the F1-score computed on the validation set $\mathcal{V}$ when using threshold $\tau$ for class $c$.

After determining the optimal thresholds for each class, we generate pseudo-labels for the unlabeled data. For each potential region in the document image $x_u$, a pseudo-label is assigned if the prediction score exceeds the corresponding optimal threshold $\tau_c^*$.

**Pseudo-Label Generation and Training** Using the optimized thresholds $\{\tau_1^*, \tau_2^*, \ldots, \tau_{23}^*\}$, the pseudo-labels $\hat{y}_u$ for the unlabeled document image $x_u$ are defined as:

$$\hat{y}_{u,i} = \begin{cases} 1, & \text{if } P(y_{i,c}|x_u) > \tau_c^* \text{ for any } c \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where $\hat{y}_{u,i}$ indicates whether the $i$-th region is assigned a pseudo-label for any category. The pseudo-labeled data,